เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 1 **Year:** 2017

Subject: 392153 Chemistry III **Section:** 15-16

Date: 27 November 2017 **Time:** 8.00 – 10.00

Name: ______________________________ ID: __________________________ Class: ________

Instructions:

1. The examination has 10 pages (including this page), 8 questions and a total score of 75 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are not allowed in the examination.
9. Electronic communication devices are NOT allowed in the examination room.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

INSTRUCTIONS: <u>Written Response</u>
Students will be expected to communicate your knowledge and understanding of chemical principles in a clear and logical manner. Students' steps and assumptions leading to a solution must be written in the spaces below the questions. Answers must include units where appropriate and be given to the correct number of significant figures. **For questions involving calculation and explanation, full marks will NOT be given for providing only an answer.**

1. Using data in the table below to determine the overall reaction order and the value of the rate constant. (10 marks)

   The chemical reaction is:

$$NH_4^+(aq) + NO_2^-(g) \longrightarrow N_2(g) + 2H_2O(l)$$

Experiment	Initial Concentration of NH_4^+ (M)	Initial Concentration of NO_2^- (M)	Initial Rate (mol/L.s)
1	0.100	0.0050	1.35×10^{-7}
2	0.100	0.010	2.70×10^{-7}
3	0.200	0.010	5.40×10^{-7}

2. The decomposition of N_2O_5 in the gas phase was studied at a constant temperature. (10 marks)

$$N_2O_5(g) \longrightarrow 4NO_2(g) + O_2(g)$$

The following results were collected:

$[N_2O]$ (mol/L)	Time (s)
0.1000	0
0.0707	50
0.0500	100
0.0250	200
0.0125	300
0.00625	400

Using these data, verify that the rate law is first order in $[N_2O_5]$, and calculate the value of the rate constant, where the rate = $\Delta[N_2O_5]/\Delta t$.

3. Butadiene reacts to form its dimer according to the equation (15 marks)

$$2C_4H_6(g) \longrightarrow C_8H_{12}(g)$$

The following data were collected for this reaction at a given temperature:

$[C_4H_6]$ (mol/L)	Time ($\pm$1 s)
0.01000	0
0.00625	1000
0.00476	1800
0.00370	2800
0.00313	3600
0.00270	4400
0.00241	5200
0.00208	6200

a. What is the order of reaction?

b. What is the value of the rate constant for the reaction?

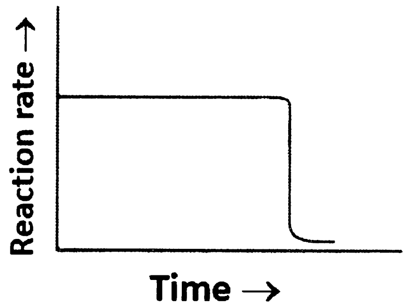
c. What is the half-life for the reaction under the conditions of this experiment?

4. The time required for 10% completion of first order reaction at 298 K is equal to that required for its 25% completion at 308K. Calculate the energy of activation when "t" is equal to (2.303/K) log (N_0/N_1) for first order reaction. (10 marks)

5. A certain first-order reaction has a half-life of 20.0 minutes. How much time is required for this reaction to be 75% complete? (10 marks)

6. Please explain which of the following graphs is correct for a zero order reaction? (5marks)

1



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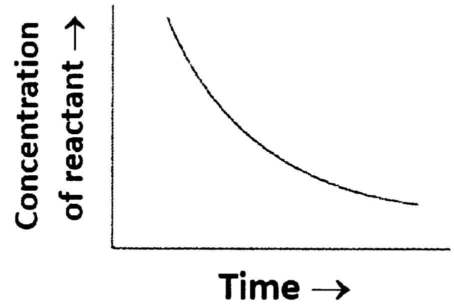

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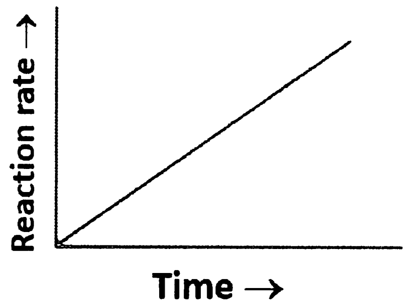

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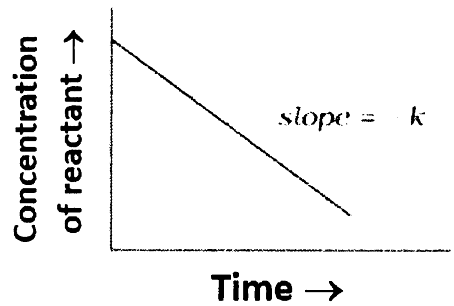

7. Describe how does the enthalpy of a reaction remain unchanged when a catalyst is used in the reaction. (5 marks)

8. At 300 K, the following reaction is found to obey the rate law:

$$\text{Rate} = k[NOCl]^2: \quad 2NOCl \rightarrow 2NO + Cl_2$$

Mechanism 1	$NOCl \rightarrow NO + Cl$	slow
	$Cl + NOCl \rightarrow NOCl_2$	fast
	$NOCl_2 + NO \rightarrow 2NO + Cl_2$	fast
	Overall: $2NOCl \rightarrow 2NO + Cl_2$	
Mechanism 2	$2NOCl \rightarrow NOCl_2 + NO$	slow
	$NOCl_2 \rightarrow NO + Cl_2$	fast
	Overall: $2NOCl \rightarrow 2NO + Cl_2$	
Mechanism 3	$NOCl \rightleftharpoons NO + Cl$	fast, equilibrium

$$NOCl + Cl \rightarrow NO + Cl_2 \qquad \text{slow}$$

$$\text{Overall: } 2NOCl \rightarrow 2NO + Cl_2$$

Consider the three postulated mechanisms given above. Then choose the response that lists all those that are possibly correct and no others.
(10 marks)

Important information

$k = A \cdot \exp(-Ea/RT)$

$R = 8.314$ J/K•mol

$\ln (k_2/k_1) = -Ea/R\ ((1/T_2) - (1/T_1))$

$$\log K_1 - \log K_2 = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

$$\ln k = -\frac{E_a}{R}\frac{1}{T} + \ln A$$

Rate Law	Concentration-Time Equation
rate = k [A]	$\ln[A] = \ln[A]_0 - kt$
rate = k	$[A] = [A]_0 - kt$
rate = k $[A]^2$	$\frac{1}{[A]} = \frac{1}{[A]_0} + kt$

$\ln 2 = 0.693$

$\ln 4 = 1.38$

$\log 1.11 = 0.046$

$\log 1.33 = 1.25$